

Name KEY

Total 12 points

Please show all your work.

1. (6 pts) The probability that an individual is left handed is 0.11. There is a group of 4 friends. Let x represent the number of individuals that are left handed.

a) (4 pts) Create a binominal probability distribution for this situation.

x		$P(x)$
0	$.89^4$	.627
1	$(.11)^1 \times (.89)^3 \times 4C_1$	.310
2	$(.11)^2 \times (.89)^2 \times 4C_2$	.058
3	$(.11)^3 \times .89 \times 4C_3$	.0047 $\leftarrow .005$ ok
4	$(.11)^4$	.0001 $\leftarrow .00015$ ok

b) (2 pts) Find the probability that at least 3 individuals (from this group) are left handed.

$$P(\geq 3) = P(3) + P(4) = .0047 + .0001 = .0048 \quad \text{or } .0051 \text{ or } .00515$$

Answer

.0048

2. (6 pts) In a pizza takeout restaurant, the following probability distribution was obtained. The random variable x represents the number of toppings for a large pizza.

Find the mean and standard deviation.

x	$P(x)$	$x - M$	$(x - M)^2$	$P(x)(x - \bar{x})^2$
0	0.28	-1.88	3.53	.372
1	0.42	-.76	.03	.011
2	0.20	.84	.71	.121
3	0.06	1.84	3.39	.203
4	0.04	2.84	8.07	.323

$$\mu = \sum xP(x) = 1.16$$

$$\sigma = \sqrt{\sum P(x)(x - \mu)^2} = \sqrt{1.059} = 1.029$$

mean

1.16

standard deviation

1.029

$\uparrow$
1.03 ok